

Math 548
Portfolio Assignment 4

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Problem 1 (F24, P1). Let G be a group, and $G \times G$ the direct product. The set $D = \{(g, g) \mid g \in G\}$ is a subgroup of $G \times G$. Prove that if D is normal in $G \times G$, then G is abelian.

Proof. Since D is normal in $G \times G$, for any $(a, b) \in G \times G$ and $(d, d) \in D$, we have:

$$\begin{aligned}(a, b)^{-1}(d, d)(a, b) \in D &\implies (a^{-1}da, b^{-1}db) \in D \\ &\implies a^{-1}da = b^{-1}db\end{aligned}$$

Take $d = b$. Then $a^{-1}ba = b^{-1}bb$, hence $ba = ab$ for any $a, b \in G$. $\square$

Problem 2 (W21, P3). Let H be a subgroup of G . The *normalizer* of H in G is the set $N_G(H) = \{g \in G \mid gH = Hg\}$.

- (a) Prove that $N_G(H)$ is a subgroup of G containing H .
- (b) Prove $N_G(H)$ is the largest subgroup of G in which H is normal.

Proof. (a) First, note that if $c \in N_G(H)$, then $cH = Hc$ implies that $Hc^{-1} = c^{-1}H$ —hence $c^{-1} \in N_G(H)$. Clearly $N_G(H)$ is nonempty as $eH = He$. If $a, b \in N_G(H)$, then:

$$\begin{aligned}(ab^{-1})H &= a(b^{-1}H) \\ &= a(Hb^{-1}) \\ &= (aH)b^{-1} \\ &= H(ab^{-1}).\end{aligned}$$

Thus $N_G(H)$ is a subgroup of G . To show that $H \subseteq N_G(H)$, it suffices to prove that $hHh^{-1} = H$. Let $h \in H$. Then $h = he = hhh^{-1} \in hHh^{-1}$. So $H \subseteq hHh^{-1}$. Let $h_xh^{-1} \in hHh^{-1}$. Since H is a subgroup of G , it is closed under products and inverses, hence $h_xh^{-1} \in H$. So $hHh^{-1} \subseteq H$. Together, $hHh^{-1} = H$, which means $H \subseteq N_G(H)$.

(b) Suppose K is another subgroup of G such that $H \trianglelefteq K$. If $x \in K$, then $xH = Hx$, hence $x \in N_G(H)$. Thus $K \subseteq N_G(H)$; i.e., $N_G(H)$ is the largest subgroup of G in which H is normal. $\square$

Problem 3 (W23, P1). Let G be a finite group and $n > 1$ an integer such that $(ab)^n = a^n b^n$ for all $a, b \in G$. Let:

$$G_n := \{c \in G \mid c^n = e\}, \quad G^n := \{c^n \mid c \in G\}.$$

You may take for granted that these are subgroups. Prove that both G_n and G^n are normal in G , and $|G^n| = [G : G_n]$.

Proof. Let $a \in G$ and $c \in G_n$. Observe that:

$$\begin{aligned} (aca^{-1})^n &= \underbrace{(aca^{-1})(aca^{-1})\dots(aca^{-1})}_{n \text{ times}} \\ &= ac^n a^{-1} \\ &= aa^{-1} \quad \text{Since } c \in G_n \\ &= e. \end{aligned}$$

Thus $G_n \trianglelefteq G$. Now let $b \in G$ and $d^n \in G_n$. Then:

$$\begin{aligned} bd^n b^{-1} &= b \cdot \underbrace{d \dots d}_{n \text{ times}} \cdot b^{-1} \\ &= \underbrace{(bdb^{-1}) \dots (bdb^{-1})}_{n \text{ times}} \\ &= (bdb^{-1})^n. \end{aligned}$$

Since $bdb^{-1} \in G$, we have $G^n \trianglelefteq G$.

Define $\varphi : G \rightarrow G^n$ by $\varphi(g) = g^n$. Then φ is a homomorphism since for any $a, b \in G$:

$$\begin{aligned} \varphi(ab) &= (ab)^n \\ &= a^n b^n \\ &= \varphi(a)\varphi(b). \end{aligned}$$

Moreover, φ is surjective since given any $g \in G^n$, we have $g = h^n$ for some $h \in G$ —whence $\varphi(h) = h^n = g$. Finally:

$$\begin{aligned} \ker \varphi &= \{g \in G \mid \varphi(g) = e\} \\ &= \{g \in G \mid g^n = e\} \\ &= G_n. \end{aligned}$$

By the First Isomorphism Theorem, $G/G_n \cong G^n$. Thus $[G : G_n] = |G^n|$. □

Problem 4 (F19, P4). Suppose G is a group, H and K are normal subgroups of G , and $H \leq K$.

- (a) Define a group homomorphism from K to G/H .
- (b) Compute the kernel of the homomorphism in (a), and apply the First Isomorphism Theorem.

Proof. (a) Define $\varphi : K \rightarrow G/H$ by $k \mapsto kH$. For any $k_1, k_2 \in K$ we have:

$$\begin{aligned} \varphi(k_1 k_2) &= (k_1 k_2)H \\ &= k_1 H k_2 H \\ &= \varphi(k_1)\varphi(k_2). \end{aligned}$$

Thus φ is a group homomorphism.

(b) We have:

$$\begin{aligned}
\ker \varphi &= \{k \in K \mid \varphi(k) = eH\} \\
&= \{k \in K \mid kH = eH\} \\
&= \{k \in K \mid k \in H\} \\
&= H.
\end{aligned}$$

By the First Isomorphism Theorem, $K/H \cong \text{im } \varphi = \{kH \mid k \in K\} = K/H$. $\square$

Problem 5. Let G be a finite group and $H, K \trianglelefteq G$ be normal subgroups of relatively prime order. Prove that G is isomorphic to a subgroup of $G/H \times G/K$.

Proof. Define $\varphi : G \rightarrow G/H \times G/K$ by $g \mapsto (gH, gK)$. Then φ is a homomorphism since for any $a, b \in G$:

$$\begin{aligned}
\varphi(ab) &= (abH, abK) \\
&= (aHbH, aKbK) \\
&= (aH, aK)(bH, bK) \\
&= \varphi(a)\varphi(b).
\end{aligned}$$

Observe that:

$$\begin{aligned}
\ker \varphi &= \{g \in G \mid \varphi(g) = (eH, eK)\} \\
&= \{g \in G \mid (gH, gK) = (eH, eK)\} \\
&= \{g \in G \mid g \in H \cap K\}.
\end{aligned}$$

Since $H \cap K$ is a subgroup of H and a subgroup of K , we must have $\text{ord}(H \cap K) \mid \text{ord}(H)$ and $\text{ord}(H \cap K) \mid \text{ord}(K)$. Thus $\text{ord}(H \cap K) \mid \gcd(\text{ord}(H), \text{ord}(K))$. But since the order of H and K are relatively prime, this means $\text{ord}(H \cap K) \mid 1$. It must be the case that $\text{ord}(H \cap K) = 1$; i.e., $H \cap K$ is the trivial group. Hence $\ker \varphi = \{e\}$. By the First Isomorphism Theorem, G is isomorphic to $\text{im } \varphi$, which is a subgroup of $G/H \times G/K$. $\square$